

SUMOTO, Japan

WMO: 477760

Lat: 34.3105N

Lon: 134.8490E

Elev: 226

StdP: 14.58

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 32.0	(c) 33.6	(d) 18.8	(e) 14.2	(f) 35.3	(g) 21.1	(h) 15.9	(i) 36.7	(j) 19.5	(k) 42.5	(l) 16.8	(m) 40.5	(n) 7.9	(o) 300	(p) 0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.8	(c) 89.6	(d) 78.1	(e) 87.8	(f) 77.6	(g) 85.8	(h) 77.0	(i) 79.7	(j) 86.6	(k) 78.7	(l) 85.1	(m) 77.9	(n) 83.7	(o) 5.3	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.7	145.3	83.5	76.9	141.2	82.6	76.1	137.5	81.8	43.5	87.1	42.4	85.2	41.6	83.9	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 17.8	(b) 14.3	(c) 12.0	(e) 28.9	(f) 92.3	(g) 1.8	(h) 1.8	(i) 27.5	(j) 93.6	(k) 26.4	(l) 94.7	(m) 25.4	(n) 95.7	(o) 24.1	(p) 97.0		
(a) 17.8	(b) 14.3	(c) 12.0	(e) 25.0	(f) 81.2	(g) 2.0	(h) 1.3	(i) 23.6	(j) 82.1	(k) 22.5	(l) 82.9	(m) 21.4	(n) 83.6	(o) 19.9	(p) 84.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.8	41.5	42.2	47.8	57.0	65.4	71.6	79.0	81.4	75.0	65.5	55.6	46.4
	DBStd	14.41	3.90	4.65	5.45	5.47	4.10	3.85	3.74	3.05	4.47	4.56	5.23	4.83
	HDD50	749	264	224	110	9	0	0	0	0	0	0	10	132
	HDD65	3101	727	639	533	245	46	2	0	0	0	48	283	578
	CDD50	4689	1	5	42	218	477	647	898	973	749	481	178	20
	CDD65	1566	0	0	0	4	58	199	433	508	299	64	1	0
	CDH74	11027	0	0	0	8	117	663	3504	4952	1685	98	0	0
	CDH80	2774	0	0	0	0	7	76	895	1501	292	3	0	0
Wind	WSAvg	5.7	6.7	6.3	6.3	5.9	5.3	4.8	5.1	5.0	5.2	5.4	5.6	6.5
Precipitation	PrecAvg	55.0	1.9	2.6	4.1	4.2	6.0	7.1	6.0	4.3	7.1	5.7	3.9	2.2
	PrecMax	81.9	4.4	6.5	7.1	8.1	16.4	14.8	12.0	14.1	14.8	16.6	11.1	5.8
	PrecMin	30.4	0.5	0.6	2.2	1.3	2.4	2.1	1.3	0.5	1.7	1.5	0.6	0.2
	PrecStd	12.7	1.1	1.5	1.3	1.9	3.1	2.9	3.3	3.8	3.7	3.7	2.6	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.9	61.5	66.3	74.7	80.5	86.0	91.8	92.6	88.9	79.7	69.9	62.7
		MCWB	51.8	57.8	56.9	61.4	65.6	75.6	79.3	78.1	77.5	72.9	65.0	58.2
	2%	DB	51.7	56.4	62.7	70.8	77.0	81.9	88.9	90.4	85.6	76.7	67.2	57.5
		MCWB	46.7	51.5	56.0	59.8	64.4	73.0	78.1	78.1	76.2	70.3	62.4	53.2
	5%	DB	49.5	52.3	59.9	68.0	74.7	79.7	86.9	88.7	83.2	74.3	65.4	55.2
		MCWB	44.6	47.2	54.0	59.5	63.9	71.5	77.6	77.8	74.9	68.4	60.7	50.4
	10%	DB	47.6	49.4	57.2	65.7	72.5	77.7	84.9	86.9	81.2	72.1	63.4	53.4
		MCWB	42.9	44.5	51.1	58.2	63.4	70.7	76.8	77.3	74.1	66.0	58.7	48.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	58.7	61.1	66.6	71.0	77.2	81.2	81.0	79.2	74.8	66.6	59.4
		MCDB	55.2	61.5	63.2	69.9	74.6	83.0	89.0	88.3	85.4	78.2	68.2	61.5
	2%	WB	48.2	52.3	57.5	63.5	68.9	75.5	79.5	79.9	77.8	71.9	63.9	54.3
		MCDB	50.2	55.6	61.2	67.4	72.5	79.7	86.5	86.9	83.0	75.5	66.2	56.9
	5%	WB	45.5	48.4	54.6	61.3	67.0	74.1	78.4	79.1	76.7	69.6	61.6	51.6
		MCDB	48.5	51.2	59.0	65.8	71.4	77.8	84.7	85.6	81.1	73.2	64.5	54.3
	10%	WB	43.5	45.3	51.9	59.3	65.3	72.5	77.4	78.3	75.6	67.3	59.4	49.4
		MCDB	47.0	48.7	56.6	64.4	70.3	76.2	83.0	84.4	79.7	71.0	62.9	52.7
Mean Daily Temperature Range	5% DB	MDBR	9.1	10.3	12.3	13.3	12.9	10.5	10.0	10.8	9.6	9.0	9.0	8.9
		MCDBR	11.4	13.5	15.6	16.0	15.5	13.0	12.4	12.8	11.5	10.2	10.1	10.6
		MCWBR	9.2	10.9	11.7	9.7	8.4	6.6	5.9	5.5	5.9	7.0	8.6	8.8
	5% WB	MCDBR	10.4	12.5	14.9	13.6	12.3	10.1	11.2	11.3	10.1	9.1	9.5	10.0
MCWBR		9.7	11.5	12.8	10.1	8.9	6.3	5.9	5.4	5.7	7.3	9.1	9.6	
Clear-Sky Solar Irradiance	taub	0.393	0.437	0.496	0.529	0.533	0.552	0.554	0.525	0.486	0.446	0.415	0.388	
	taud	2.196	2.086	1.939	1.912	1.972	2.002	2.047	2.119	2.199	2.225	2.218	2.225	
	Ebn at Noon	250	251	246	245	245	240	239	243	247	246	241	244	
	Edh at Noon	36	45	56	61	58	56	53	49	43	39	36	34	
All-Sky Solar Radiation	RadAvg	784	976	1284	1538	1658	1485	1618	1668	1298	1057	815	705	
	RadStd	55	74	93	142	164	144	202	152	114	106	69	44	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+1.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page